```
Queue Menu
1. Insert an element
2. Delete an element
3. Display the queue
4. Exit the program
Enter your choice: 5
Invalid choice! Please select between 1 and 4.

Queue Menu
1. Insert an element
2. Delete an element
3. Display the queue
4. Exit the program
Enter your choice: 1
Enter the value to insert: 1

Queue Menu
1. Insert an element
2. Delete an element
3. Display the queue
4. Exit the program
Enter your choice: 1
Enter the value to insert: 2

Queue Menu
1. Insert an element
2. Delete an element
3. Display the queue
4. Exit the program
Enter your choice: 1
Enter the value to insert: 3

Queue Menu
1. Insert an element
2. Delete an element
3. Display the queue
4. Exit the program
Enter your choice: 1
Enter the value to insert: 4

Queue Menu
1. Insert an element
2. Delete an element
3. Display the queue
4. Exit the program
Enter your choice: 1
Enter the value to insert: 5
```

```
Queue Menu
1. Insert an element
2. Delete an element
3. Display the queue
4. Exit the program
Enter your choice: 1
Enter the value to insert: 6
Queue Overflow (The Queue is full, cannot enter more elements)

Queue Menu
1. Insert an element
2. Delete an element
3. Display the queue
4. Exit the program
Enter your choice: 3
Queue elements: 1 2 3 4 5

Queue Menu
1. Insert an element
2. Delete an element
3. Display the queue
4. Exit the program
Enter your choice: 2
Deleted element: 1

Queue Menu
1. Insert an element
2. Delete an element
3. Display the queue
4. Exit the program
Enter your choice: 2
Deleted element: 2

Queue Menu
1. Insert an element
2. Delete an element
3. Display the queue
4. Exit the program
Enter your choice: 2
Deleted element: 3

Queue Menu
1. Insert an element
2. Delete an element
3. Display the queue
4. Exit the program
Enter your choice: 2
Deleted element: 4
```

```
Queue Menu
1. Insert an element
2. Delete an element
3. Display the queue
4. Exit the program
Enter your choice: 2
Deleted element: 5

Queue Menu
1. Insert an element
2. Delete an element
3. Display the queue
4. Exit the program
Enter your choice: 2
Queue is empty, nothing to delete

Queue Menu
1. Insert an element
2. Delete an element
3. Display the queue
4. Exit the program
Enter your choice: 3
Queue is empty, nothing to display.

Queue Menu
1. Insert an element
2. Delete an element
3. Display the queue
4. Exit the program
Enter your choice: 4
You have exited the program.
Bye!
```

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8) Write a Program to simulate the working of a queue of integers using an array. Provide the following operations: Insert, Delete, Display. The program should print appropriate messages for queue empty and queue overflow conditions.

Pseudocode:-

```
DEFINE MAX = 5
DECLARE queue [MAX]
DECLARE front = -1, rear = -1

FUNCTION enque ~~insert~~ ()
   IF (rear == MAX - 1) THEN
      PRINT "QUEUE Overflow"
   ELSE
      Read item
      IF (front == -1) THEN
         front ← 0
      ENDIF
      rear ← rear + 1
      queue [rear] ← item
   ENDIF
END FUNCTION

FUNCTION deque ~~delete~~ ()
{ if (front == -1 && rear == -1) {printf("Empty queue")
  else if (front == rear)
  { front = rear = -1; }
  else {
  print ("% ", queue [front] ) }
  front ++; }
```

```
void display(){
  if (front = = -1&&rear = =-1){printf("Empty queue")}
else { for (int = front ; i<=rear; i++)
    {print("%d", queue[i])}}
  }
```

~~O/P~~

```
#include <stdio.h>
#define N 5

int queue[N];
int front = -1;
int rear = -1;

void enque(int x)
{
    if(rear == N-1)
    {
        printf("Queue Overflow(The queue is full,
         cannot enter more elements\n");
    }
    else if (front == -1 && rear == -1)
    {
        front = rear = 0;
        queue[rear] = x;
    }
    else
    {
        rear ++;
        queue[rear] = x;
    }
}
```

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```
void deque()
{
    if(front == -1 && rear == -1)
    {
        printf("Queue is empty, nothing to delete\n");
    }
    else if(front == rear)
    {
        printf("Deleted element: %d\n", queue[front]);
        front = rear = -1;
    }
    else
    {
        printf("Deleted element: %d\n", queue[front]);
        front++;
    }
}

void display()
{
    if (front == -1 && rear == -1)
    {
        printf("Queue is empty, nothing to display\n");
    }
    else
    {
        printf("Queue elements: ");
        for (int i= front; i<= rear; i++)
        {
            printf("%d", queue[i]);
        }
        printf("\n");
    }
}
```

```c
int main()
{
    int choice, value;
    while (1)
    {
        printf("\nQueue Menu\n");
        printf("1. Insert an element\n");
        printf("2. Delete an element\n");
        printf("3. Display the queue\n");
        printf("4. Exit the program\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
        case 1 :
            printf("Enter the value to insert: ");
            scanf("%d", &value);
            enque (value);
            break;
        case 2 :
            deque();
            break;
        case 3 :
            display ();
            break;
        case 4 :
            printf("You have exited the program.
            \n Bye!\n");
            return 0;
        default :
            printf("Invalid choice! Please select
            between 1 and 4.\n");
        }
    }
```

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O/P Queue menu
1. Insert an element
2. Delete an element
3. Display the queue
4. Exit the program
Enter your choice: 5
Invalid choice! Please select between 1 and 4.

Elements : 1, 2, 3, 4, 5

→ Queue Overflow (The Queue is full, cannot enter more elements)
→ Queue elements : 1 2 3 4 5
→ Queue is empty, nothing to delete
→ Queue is empty, nothing to display.
→ You have exited the program.
Bye!

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